

JKBOSE 2026

CLASS 11TH

#XUV 

EQUILIBRIUM

LECTURE-1 CHEMISTRY

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ON
TOP



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Topics To Covered

- 1** Physical Equilibrium
- 2** Chemical Equilibrium

REVISION

ENTROPY.

i) For Spontaneous Process

$$\Delta S = +ve, \Delta G = -ve$$

ii) For Non-Spontaneous process

$$\Delta S = -ve, \Delta G = +ve$$

iii) For Reversible or Equilibrium

$$\Delta S = 0, \Delta G = 0$$

GIBBS FREE ENERGY.

$$G = H - TS$$

$$\Delta G = \Delta H - T\Delta S.$$

The End

Relationship between ΔG° and Equilibrium Constant (K_{eq}).

ΔG & ΔG° are related to each other by

$$\Delta G = \Delta G^\circ + RT \ln Q.$$

At equilibrium, $\Delta G = 0$, Q (Reaction Quotient) = K_{eq} .

$$0 = \Delta G^\circ + RT \ln K_{eq}$$

$$\Delta G^\circ = -RT \ln K_{eq}.$$

Thermodynamics

END.

UNIT-VI: EQUILIBRIUM

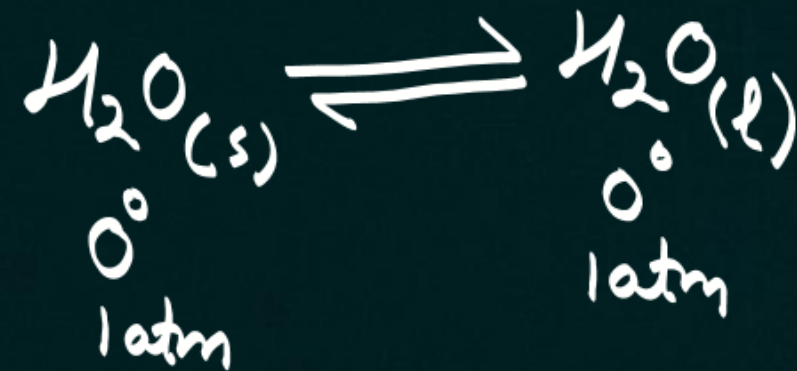
07 MARKS

- Equilibrium in physical and chemical processes, dynamic nature of equilibrium, law of mass action, equilibrium constant, factors affecting equilibrium: Le-Chatelier's principle equilibrium-ionization of acids and bases, strong and weak electrolytes, degree of ionization, Concept of pH. Hydrolysis of salts (elementary idea), buffer solutions. Solubility product, common ion effect (with suitable examples).

#EQUILIBRIUM.

It is defined as stage of a process at which pressure, temperature and concentration do not change with the passage of time.

e.g melting of ice



Two types of Equilibrium

1) Physical Equilibrium

2) Chemical Equilibrium

1) Physical Equilibrium

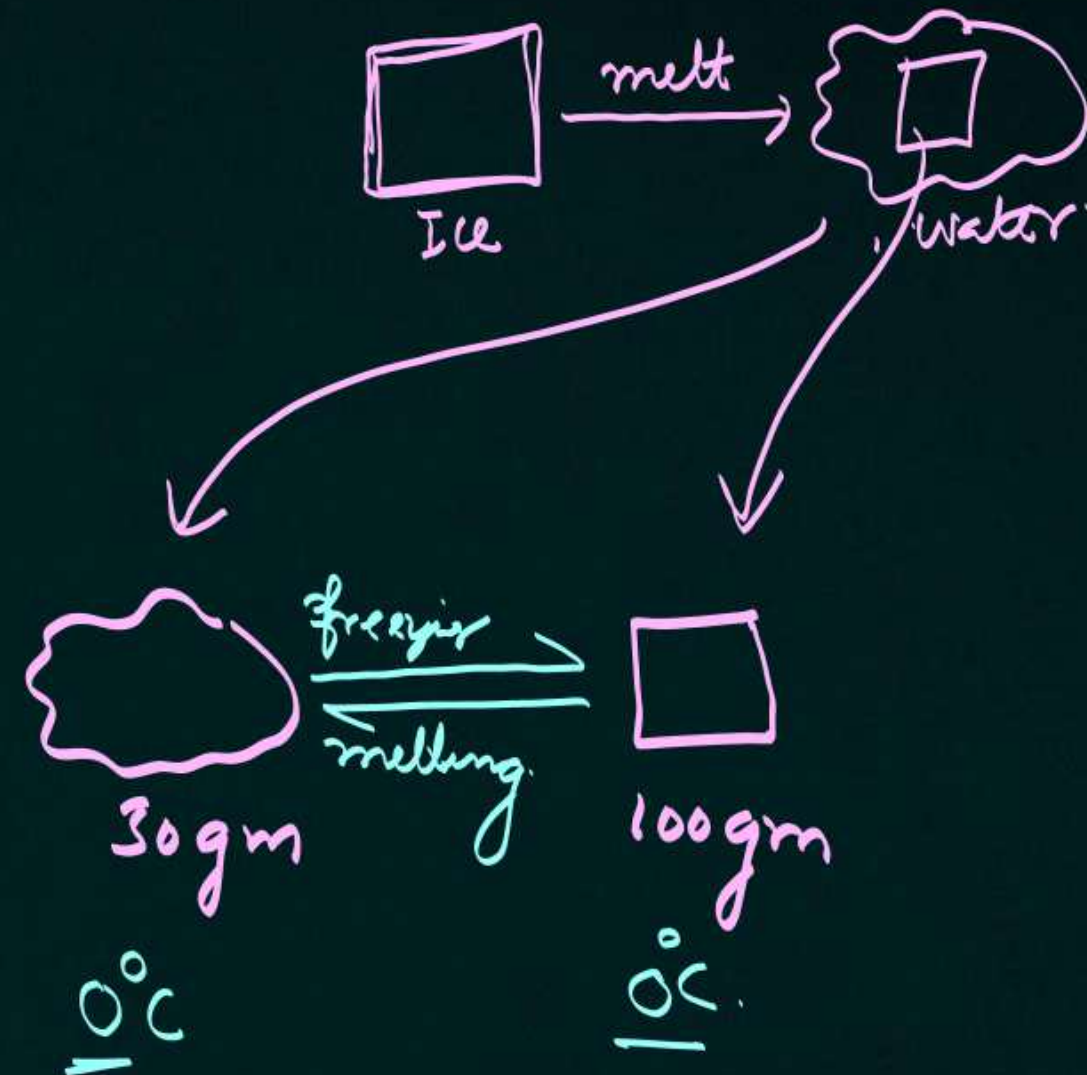
When equilibrium is present in physical change then it is known as physical equilibrium.

Five type

- Solid-liquid Equilibrium
- Liquid-Vapour Equilibrium
- Solid-Vapour Equilibrium
- Solid-Solution Equilibrium
- Gas - solution Equilibrium

Physical Equilibrium

1) Solid-liquid $\rightarrow$ Let us take the example of piece of ice and allow it to melt and then measure the weight of ice and water, then put it in a thermos and maintain the temp. at 0°C . After some time again measure the weight of ice and water of thermos. We found that there is no change in weight of ice and water. This is because, there exist a equilibrium between liquid & ice.



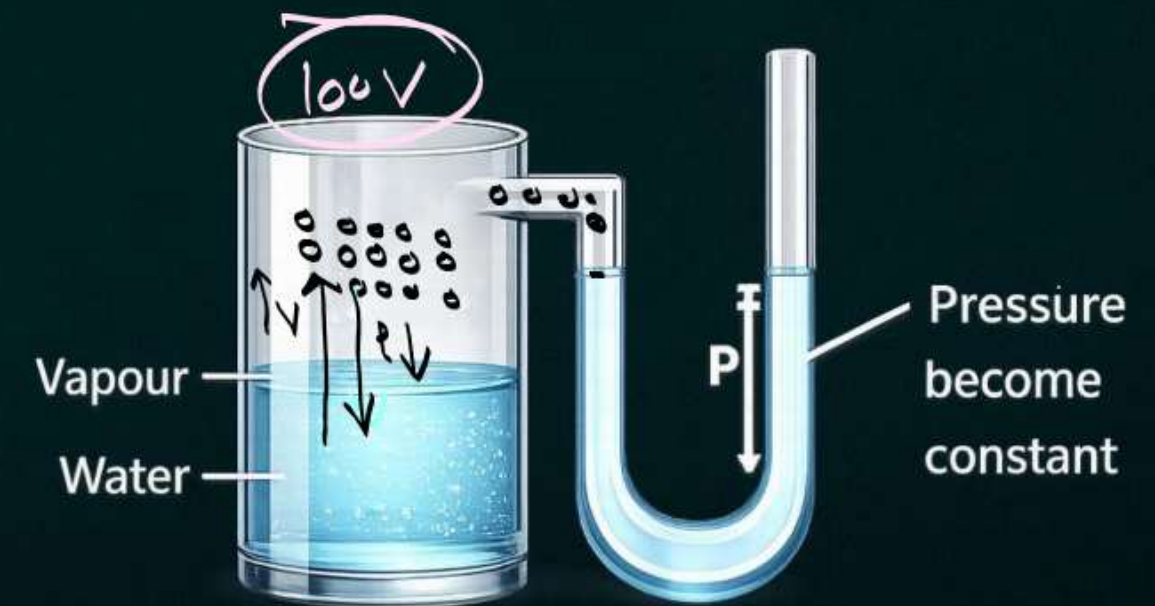
Equilibrium in Physical Processes

2. Liquid Vapour Equilibrium:

- When liquid water is placed in a closed container, it starts evaporating, increasing vapour pressure. This is shown by a rise in mercury level in a manometer. Eventually, a constant pressure is reached, indicating equilibrium.
- At the molecular level, evaporation increases vapour molecules, which collide with the liquid and condense. Initially, evaporation is faster than condensation, but over time, both rates become equal. This balance maintains constant pressure - the state of liquid-vapour equilibrium.

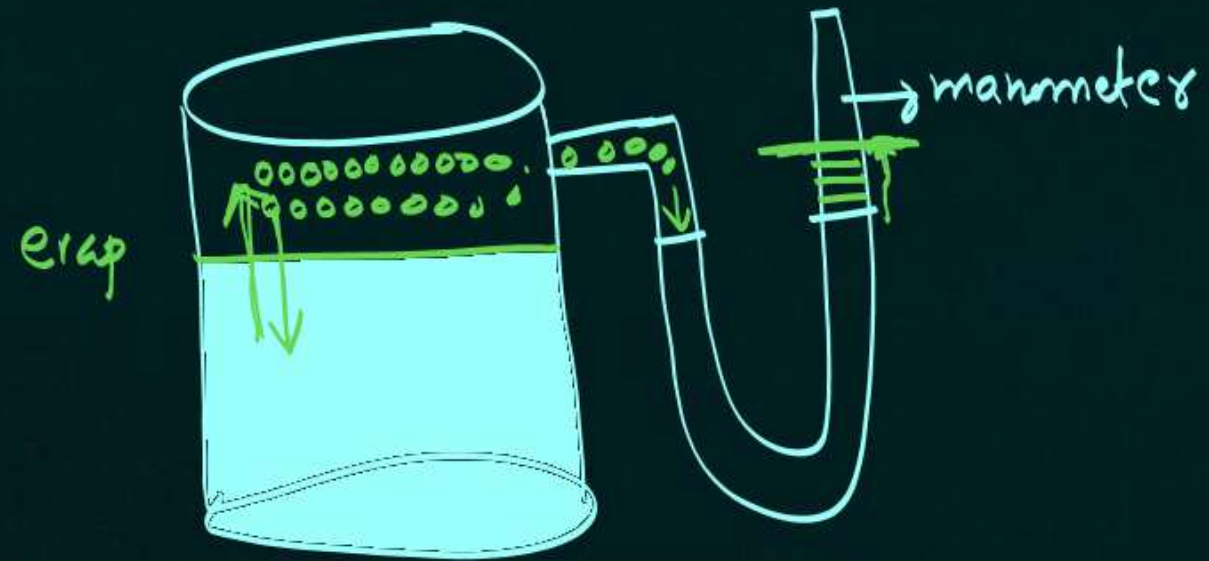


Initial stage (a)



Equilibrium stage (b)

Rate of evaporation = Rate of condensation



Physical Equilibrium



3. Solid – Vapour Equilibrium:

Let us take the piece of iodine in a closed transparent container, eventually iodine sublimes and violet colour will start appearing inside the container but after some time intensity of violet colour will not change. At this stages there will be equilibrium between.

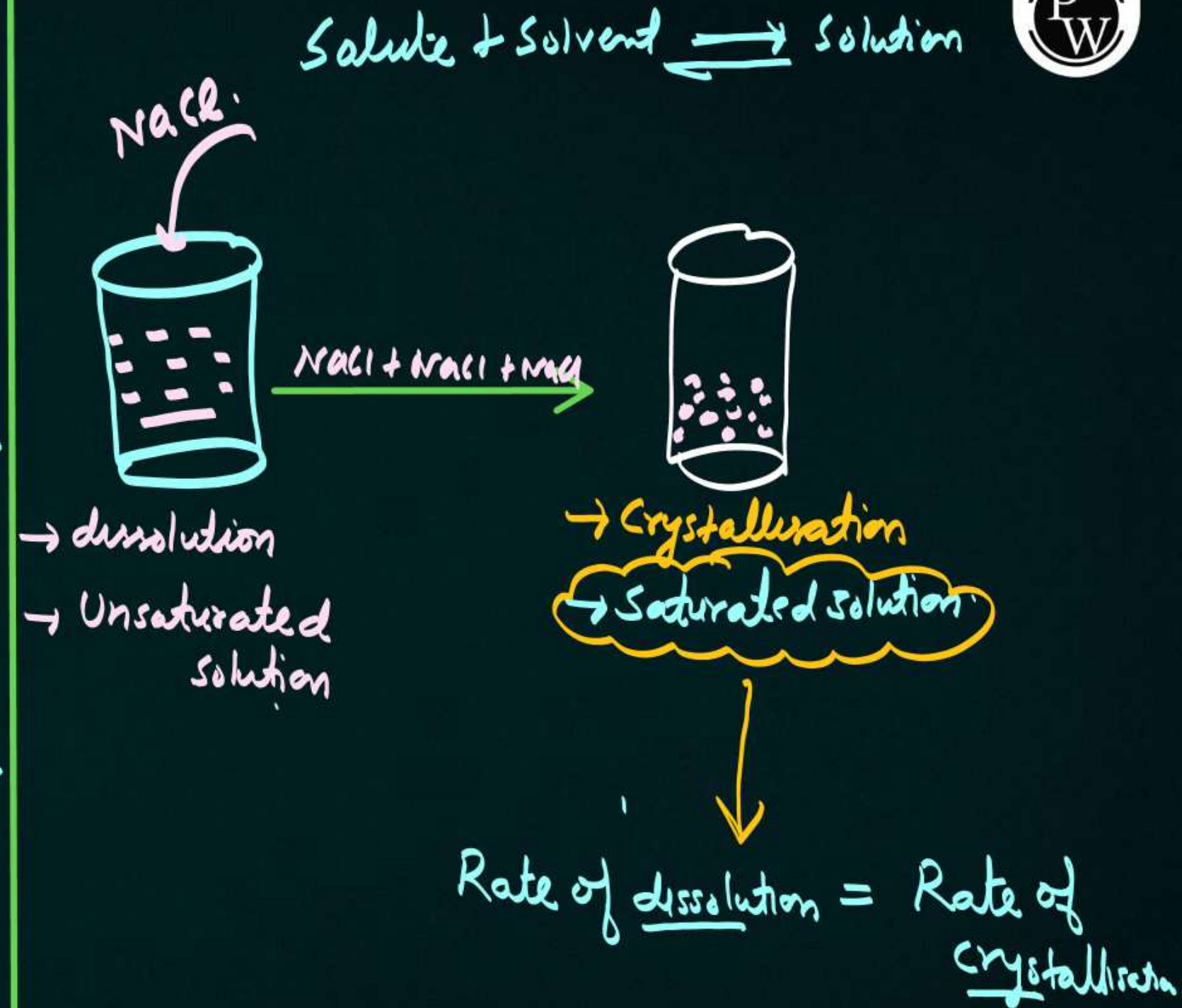


d) Solid-Solution Equilibrium

Let us dissolve NaCl (solute) in water, it gets easily dissolved & process is known as dissolution. This type of solution in which more solute can be dissolved is known as unsaturated solution. But if we keep adding NaCl in water continuously, the stage will reach, NaCl crystals will start appearing in solution. This process is known as crystallisation. That means no more NaCl can be dissolved, this type of solution in no more solute can be dissolved is called. Saturated solution. **At this stage.**



there is equilibrium between solid & solution.



Physical Equilibrium



5. Gas – Solution Equilibrium:

Let us take the example of carbonated drink.

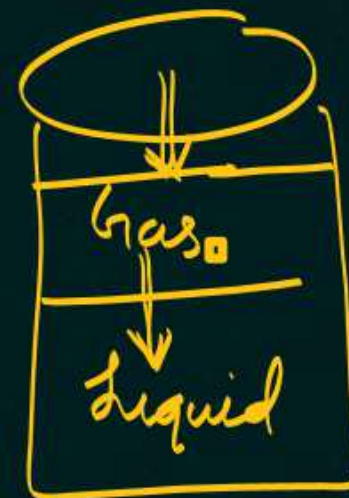
When we open the bottle of any cold drink, the gas present inside it fizzes out with the pressure.

There exist a equilibrium between gas & solution



Dissolution gas in liquid follows Henry law, it states that mass of gas dissolved in liquid is directly proportional to the pressure of the gas above the surface of liquid"

$$m \propto P$$



QUESTION



Match the column I with column II and mark the appropriate choice.

Column - I		Column - II	
(A)	Liquid $\rightleftharpoons$ Vapour	(i)	Saturated solution
(B)	Solid $\rightleftharpoons$ Liquid	(ii)	Boiling point
(C)	Solid $\rightleftharpoons$ Vapour	(iii)	Sublimation point
(D)	Solute _(s) $\rightleftharpoons$ Solution _(solution)	(iv)	Melting point

1

(A) $\rightarrow$ (i), (B) $\rightarrow$ (iii), (C) $\rightarrow$ (ii), (D) $\rightarrow$ (iv)

2

(A) $\rightarrow$ (ii), (B) $\rightarrow$ (iv), (C) $\rightarrow$ (iii), (D) $\rightarrow$ (i)

3

(A) $\rightarrow$ (iv), (B) $\rightarrow$ (ii), (C) $\rightarrow$ (i), (D) $\rightarrow$ (iii)

4

(A) $\rightarrow$ (iii), (B) $\rightarrow$ (iv), (C) $\rightarrow$ (ii), (D) $\rightarrow$ (i)

QUESTION

The equilibrium between water and its vapour, in an open vessel.

- 1 Can be achieved
- 2 Depends upon pressure
- 3 Cannot be achieved
- 4 Depends upon temperature.



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③ Muraad

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REVISE Kar lena

Thank #JIGAR
You